MIL-M-60065(MU)
11 May 1964
Superseding
TL-PD-23
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MILITARY SPECIFICATION
FOR
MOTOR, BELLOWS, EXPLOSIVE, M6

1. SCOPE

1.1 This specification covers one type of electrically-initiated, gas-actuated, one-time operation, carbon bridge bellows motor.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550 -- Ammunition and Special Weapons, General Specification for
MIL-I-45208A -- Inspection System Requirements with Appendix A, Quality Assurance System

STANDARDS

Federal

Fed. Test Method Std. No. 406 -- Plastics: Methods of Testing, Method 7021

Military

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 -- Marking for Shipment and Storage

/FSC 1325/

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DRAWINGS

Army Materiel Command

EPL 10401508 -- Motor, Bellows, Explosive, M6

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49 CFR Parts 71 to 90 -- Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite 49 CFR Parts 71 to 90 (latest revision)).

2.3 Precedence of documents. In the event of conflict between the requirements of this specification and other applicable documents, the order of precedence shall be as follows:

a. Contract
b. Specification
c. Drawings
d. Detail specifications for materials or operation
e. General specifications to classes of materials
f. Test equipment manuals

3. REQUIREMENTS

3.1 Initial production sample. Prior to the start of regular production except where a new contract follows within 90 days the production of acceptable material from the same production line, the contractor shall manufacture and submit a sample of 350 motors and ten sets of parts, using as far as possible the methods and processes proposed for quantity production. However, any deviation from these methods and processes shall be documented and submitted with the initial production sample. The sample

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shall conform to the requirements of this specification. Additional samples that may be required because of the failure of the sample to meet the requirements of this specification shall be supplied by the contractor at his expense. Until the initial production sample is approved further production will be at the risk of the contractor and the Government will not proceed with the inspection for acceptance of production lots until this approval has been obtained. Any initial production sample test conducted at the contractor's plant must be witnessed by a qualified representative of the cognizant Government technical laboratory.

3.1.1 Supplemental sample. Prior approval of the procuring activity shall be required whenever there is a change in design, material, or process which affects safety, operability, or interchangeability of any part of the assembly. The decision for requesting a supplemental sample shall rest with the technical laboratory. If the Government directs the contractor to make a change, the Government will bear the cost of the sample and all necessary inspection and testing. If the contractor requests the change, he shall bear all the costs and the Government only the cost of testing performed by the Government.

3.2 Materials, design, and construction. The motor shall be in accordance with this specification.

3.3 Performance

3.3.1 Bridge resistance. The resistance of the bridge shall be from 750 ohms to 15,000 ohms.

3.3.2 Lead fatigue. There shall be no mechanical damage and the requirements of 3.3.1 shall be met after each motor lead is subjected to an 8-pound pull.

3.3.3 Centrifuge. The motor shall meet the requirements of 3.3.1 and 3.3.4 after subjection to an acceleration of 20,000g in a centrifuge for five seconds in both directions along the logitudinal axis of the motor and in the transverse direction.

3.3.4 Functioning

3.3.4.1 Sensitivity. The motor shall function and provide movement against a load, (see 3.3.4.2), when subjected to the discharge of a 0.04 microfarad capacitor ± 1 percent charged to 55 volts. (The term movement, as used in this paragraph shall mean any measureable elongation of the motor bellows greater than 0.030 inch).

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3.3.4.2 Motor. Upon actuation, the motor mounted in a 15/16 inch (outside diameter) radial recess, 5/16 inch wide shall provide a minimum movement of 60 degrees against a load, increasing from two pounds initially to ten pounds at the end of travel without bursting or cracking.

3.3.5 Vibration. There shall be no mechanical damage affecting functioning and the requirements of 3.3.4 shall be met when the motor is tested following subjection to vibration having a double amplitude of 0.06 inch and a frequency varying uniformly from 10 cycles per second to 55 cycles per second and back to 10 cycles per second in one minute, for a period of 15 minutes in each and any direction (see 4.7.5).

3.3.6 Temperature cycling. The motor shall meet the requirements of 3.3.4 following subjection to temperature cycling between the limits of -65°F and 160°F.

3.3.7 Low temperature. The motor shall meet the requirements of 3.3.4 when the temperature of the motor is -65°F.

3.3.8 High temperature. The motor shall meet the requirements of 3.3.4 when the temperature of the motor is 160°F.

3.3.9 Forty-foot drop. The supported motor shall be subjected to a 40-foot drop onto a 3-inch thick steel plate. The motor shall not have fired.

3.4 Workmanship. All parts shall be manufactured and finished in a thoroughly workmanlike manner to insure satisfactory functioning and durability after extended periods of storage and extreme conditions encountered in service use. All parts shall be free from burrs, chips, dirt, grease, and other foreign material. Cleaning operations shall be in accordance with best commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor quality assurance system. The contractor shall provide and maintain an adequate quality assurance system in compliance with specification MIL-I-45208A w/appendix A.

4.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification in compliance with specification MIL-I-45208A w/appendix A.

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4.3 Initial production sample. The first 350 motors and ten sets of inert parts on any new contract (except where a new contract follows production of acceptable material of identical design) shall be inspected and tested by the technical laboratory in accordance with the flow chart for initial production sample of figure 1.

4.3.1 Acceptance of initial production sample. If the sample passes the criteria on the flow chart, it shall be approved. Upon failure of the initial production sample in any of the inspections and tests, the results of the inspection shall be analyzed by the technical laboratory. These results together with the engineering analysis of the motors delivered shall form a basis for corrective action. Depending on the degree of corrective action deemed necessary by the technical laboratory, the initial production sample may be:

(a) Accepted, provided recommended changes are made by the contractor prior to the start of regular production.

(b) Rejected, and new samples for approval required for tests failed.

(c) Rejected, and a complete new initial production sample required for approval.

In all cases, the contractor shall control any recommended changes to the satisfaction of the Government in future production.

4.4 Control sample. Ten motors, selected at random from each accepted lot or selected as specified by the technical laboratory, shall be shipped within two days to the technical laboratory for inspection. The motors will be evaluated by the technical laboratory. Results of the evaluation will not affect lot acceptance, but may form a basis for corrective action by mandatory changes in subsequent lots.

4.5 Acceptance inspection. Inspection shall be as specified in MIL-A-2550 and as specified herein. The inspection under Classification of Defects (4.5.5) and Acceptance Tests (4.5.6) shall be conducted by the contractor under the observation of the Government inspector, unless otherwise directed by the procuring activity.

4.5.1 Lot formation. Lots of motors shall consist of units manufactured of the same materials under essentially the same conditions and shall not exceed 10,000 units per lot. The primer components and propellant components for a lot of motors shall be limited to a single lot of each.

4.5.2 Sampling. Sampling shall be in accordance with Standard MIL-STD-105.

4.5.3 Lot acceptance. Lots shall be accepted on the basis of Classification of Defects (4.5.5), Acceptance Tests (4.5.6), and Certification (4.5.7).

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4.5.4 Data recording. A written record shall be made of all inspections and tests as specified in this specification. The data shall be recorded on forms approved by the Government agency. The forms properly filled out shall be forwarded to the cognizant Government agency after Government acceptance of the lot.

4.5.5 Classification of defects

4.5.5.1 Final assembly (see drawing C10401508)

Categories	Defects	Method of inspection
Major	AQL 1.0 percent	
101.	Outside diameter of bellows (maximum)	Gage
102.	Length (maximum)	Gage
103.	Sealing	DS-1
104.	Solder joint 360° (if solder joint is used)	Visual
105.	Lead wires not properly shorted	Visual
106.	Leads damaged	DS-2
107.	Housing damaged	DS-3
108.	Marking	DS-4

4.5.5.2 Defects standards

DS-1 There shall be continuous bonding and no opening, cracks, or uncovered areas penetrating through the seal.

DS-2 The first two inches of each lead adjacent to the motor shall be free from all kinks, nicks (more than 1/3 the lead diameter in depth), breaks, twists, and sharp bends.

DS-3 There shall be no tears in the crimping folds. Non-functional dents shall not be a cause for rejection.

DS-4 Marking shall be as specified on the applicable documents.

4.5.5.3 Preparation for delivery. By observation of the contractor's packing procedures, the inspector shall ascertain that the requirements of section 5 are met.

4.5.6 Acceptance tests. All tests specified in table I shall be performed in the order listed. The tests shall be performed on sample motors passing the Classification of Defects inspection, unless the Government considers it more practical to select a separate sample from the lot. Motors subjected to the Centrifuge test shall not be returned to the lot.

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TABLE I

Test	Requirement Par.	Test Par.	Classification of Paragraph	AQL (percent defective)
a. Bridge resistance	3.3.1	4.7.1	Test	0.65
b. Lead fatigue	3.3.2	4.7.2	Test	1.00
c. Centrifuge	3.3.3	4.7.3	Conditioning	
(1) Bridge resistance	3.3.1	4.7.1	Test	1.00
(2) Sensitivity	3.3.4.1	4.7.4	Test	0.65
(3) Motor	3.3.4.2	4.7.4	Test	*2.5

* Sensitivity failures shall be included as failures under the 2.5 AQL for the Motor test.

4.5.7 Certification.

4.5.7.1 Acetone extract of molded phenolic resin material. The contractor shall certify that at least one test has been made on each lot of molded phenolic resin parts, and on each batch of new material, and also, whenever a change is made in the molding or curing process. The Acetone extract test shall be as prescribed in Fed. Test Method Std. No. 406, Method 7021 (see drawing C10401512.

4.5.7.2 Sleeve and bridge assembly. The contractor shall certify that the force required to seat the sleeve on the bridge assembly conforms to the requirement of drawing B10401532.

4.5.7.3 Insulation resistance. The contractor shall certify that the insulation resistance of the plug and lead wire assembly conforms to the requirement of drawing C10401512.

4.6 Test Conditions and equipment.

4.6.1 Test conditions. Unless otherwise specified herein all measurements and tests shall be made in the temperature range of 65°F to 95°F ambient atmospheric pressure and ambient relative humidity. Where a temperature is specified, it shall be maintained within ±4°F. All voltage shall be held to ±1 percent.

4.6.2 Test equipment. Test equipment shall be of sufficient accuracy and quality to permit performance of the required inspection. The manufacturer shall establish and maintain adequate calibration of test equipment to the satisfaction of the Government. Meters shall be accurate within two percent and, when practicable, shall be marked conspicuously to indicate the limit of acceptance for each test. The quality, accuracy, and calibration of the test equipment shall be approved by the procuring activity. Prior to approval, the contractor shall submit to the procuring activity written test and calibration procedures.

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4.7 Test procedures

4.7.1 Bridge resistance. The resistance of the bridge shall be measured between the lead wires, using a current not to exceed 50 microamperes. (See 3.3.1 and 6.2).

4.7.2 Lead fatigue. The motor shall be held fixed and a tensile load shall be gradually applied to each lead in the direction of the longitudinal axis of the motor until the applied load is 8 pounds. This load shall be applied for a period of 30 seconds and the 8-pound limit shall be reached after a period of gradual application in excess of one second (see 3.3.2).

4.7.3 Centrifuge. The motor shall be mounted in a centrifuge so that the longitudinal axis of the motor lies parallel to the radial arm of the centrifuge. While in that position the motor shall be spun as rapidly as the equipment permits to an acceleration of 20,000 and held at that value for a minimum of five seconds. The procedure shall be repeated with the motor in the reverse position, and with the motor in the transverse position (see 3.3.3).

4.7.4 Functioning. The motor shall be confined in a 15/16 inch (outside diameter) radial recess 5/16 inch wide against a load exerting an initial force of two pounds and having a potential of exerting a force greater than ten pounds. A minimum movement of 60 degrees against the load shall result upon initiation of the motor by discharging a 0.04 microfarad capacitor charged to 55 volts across the motor leads. The extension of the bellows shall be measured (3.3.4).

4.7.5 Vibration. The motor shall be securely held and vibrated with a frequency varying from 10 cycles per second to 55 cycles per second and back to 10 cycles in one minute, for 15 minutes. The double amplitude of the vibration shall be 0.06 inch. The motor shall be oriented in each of the following directions:

1. Longitudinal axis perpendicular to the direction of vibration
2. Longitudinal axis parallel to the direction of vibration

The motor shall then be examined for possible mechanical damage affecting functioning and tested in accordance with 4.7.4 (see 3.3.5).

4.7.6 Temperature cycling. The motor shall be subjected to not less than three complete and uninterrupted temperature cycles between the limits of -65°F and 160°F, as illustrated in figure 2. After completion of cycling, the motor shall be removed from the chamber and allowed to return to normal temperature, and then tested in accordance with 4.7.4 (see 3.3.6).

Note: Single shift operation is permissible by lengthening the duration of the conditioning at either temperature. The 47-hour range may be extended accordingly.

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4.7.7 Low temperature. The motor shall be subjected to an ambient temperature of -65°F for not less than two hours. The motor shall be removed from the cold box and within one minute shall be tested in accordance with 4.7.4 (see 3.3.7).

4.7.8 High temperature. The motor shall be subjected to an ambient temperature of 160°F for not less than two hours. The motor shall be removed from the hot box and within one minute shall be tested in accordance with 4.7.4 (see 3.3.8).

4.7.9 Forty-foot drop. Motors shall be mounted in a plastic housing so formed as to hold (1) axes of one-half of the motors normal to the axis of the housing and (2) the remaining motors with their axes parallel to the axis of the housing. The parallel motors shall be oriented to produce a test assembly in which the leads are nearest the forward end of the adapter. The plastic housing shall then be assembled to an adapter and subjected to a forty-foot drop test. The adapter shall be dropped in such a manner that the aft end of the adapter will strike the armor plate. The motor shall be examined to determine if firing has occurred (see 3.3.9).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Preservation and packaging shall be level C and shall be in accordance with the manufacturer's commercial practice.

5.1.2 The motors shall be completely wrapped in aluminum foil 0.001 inch thick minimum (MIL-A-148). The contractors shall meet the requirements of 49 CFR Parts 71-90, Explosives and other Dangerous Articles for packaging.

5.2 Marking. In addition to any special marking required by the contract or order, package and shipping containers shall conform to the requirements of MIL-STD-129 and 49 CFR Parts 71-90 for the shipment of explosives and other dangerous articles. Lot, batch or control number shall be included in unit packaging, box loading, and packing operations. In addition, each foil wrapped package shall bear a card with the following notation legibly printed on both sides: CAUTION ELECTRICALLY GROUND THIS PACKAGE BEFORE OPENING. HANDLE ONLY UNDER FULLY GROUNDED CONDITIONS AND IN APPROVED LOCATIONS.

5.3 Data cards. Data card information shall be as specified in MIL-A-2550.

6. NOTES

6.1 Procurement data

6.1.1 Ordering data. Procurement documents should specify the following: Title, Number, and Date of this specification.

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6.1.2 Samples. The number of motors required for delivery should also include samples as required in 4.3, 4.4, and 4.5.6.

6.1.3 Lead time. The testing of the initial production sample will require a maximum of 20 working days. If there are failures, additional time will be required.

6.2 Advisory test equipment manuals

6.2.1 NAVORD OD 6820 Test Set Mark 162 Mod 1, A Primer Resistance Test Set for Electric Primers Used in the Proximity (VT) Fuze.

6.2.2 NAVORD OD 6818 Test Set Mark 160 Mod 0, an Insulation Resistance Test Set for Electric Primers Used in the Proximity (VT) Fuze.

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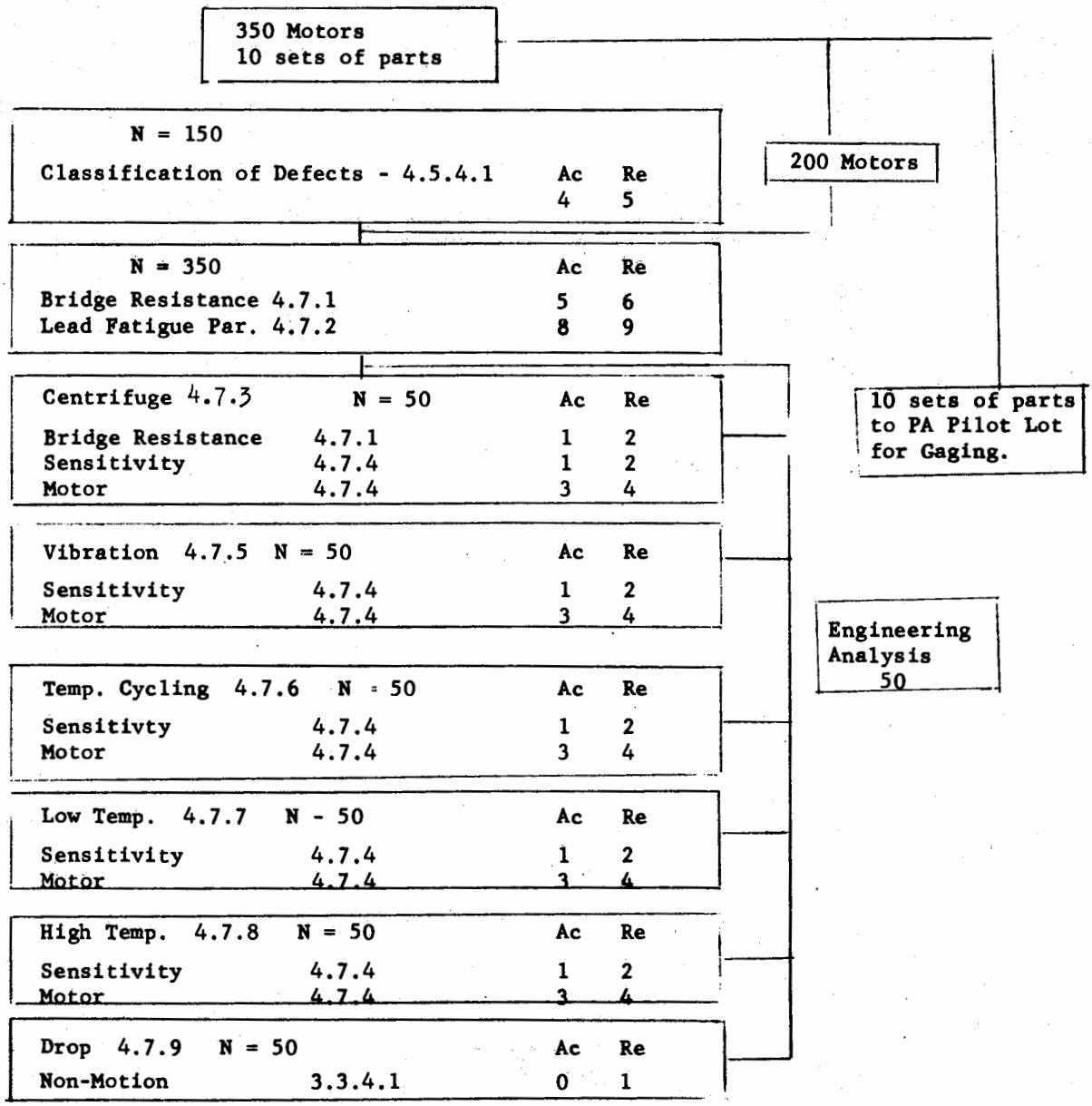
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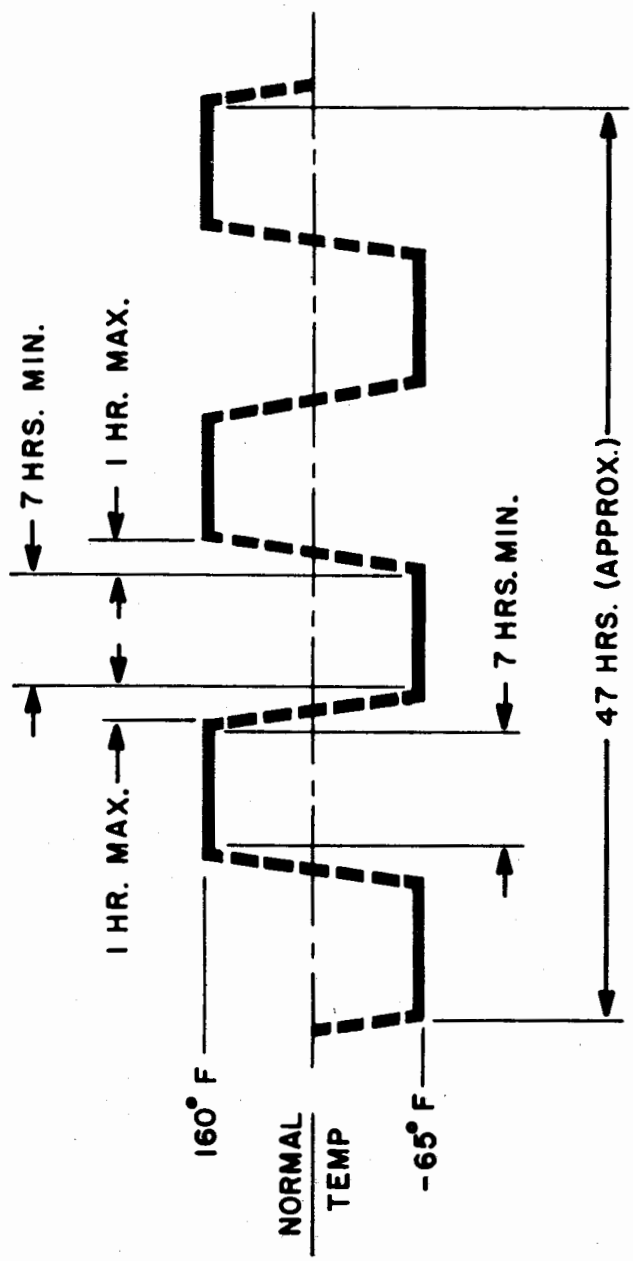
M6 INITIAL PRODUCTION SAMPLE FLOW CHART



All initial production sample tests are to be performed by the Technical Laboratory

Figure 1

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THE STRAIGHT LINE CHANGE FROM ONE TEMPERATURE TO ANOTHER IS NOT INTENDED TO IMPLY THAT THIS CONVERSION MUST BE MADE AT A UNIFORM RATE.

FIGURE 2

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